

GBM Part I: mathematical theory and experimental design

Direction ④ · Geometric Brownian motion benchmark · Euler-Maruyama and Milstein

This handout derives the four questions in the Problem Pack. It distinguishes exact mathematical quantities from estimates obtained using finitely many Monte Carlo paths. The optional positivity study is included.

Question	Mathematical result	What to show in the report
1. Strong and weak convergence	Pathwise error and expectation bias; predicted rates	Two log-log plots, fitted slopes, confidence intervals for weak bias
2. Terminal distribution	Exact lognormal law of $S(T)$	Numerical histogram and exact density; quantiles
3. Moment checks	Exact mean and variance	Small comparison table for simulated and exact moments
4. Positivity (optional)	When one EM or Milstein factor is negative	Rate of negative paths versus h ; interpretation

0 · Set up the model and numerical methods

Assume $S_0 > 0$, $T > 0$, constant drift μ , and constant volatility $\sigma > 0$. Let $t_n = nh$, $h = T/N$, where N is an integer. Brownian increments for path m are independent in both m and n :

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad \Delta W_n^{(m)} = \sqrt{h} Z_n^{(m)}, \quad Z_n^{(m)} \sim N(0, 1)$$

Use the same increment at step n for EM, Milstein, and the exact solution on a given path. Repeat independently for $m = 1, \dots, M$; for cross-resolution comparisons, construct coarse increments by summing corresponding fine increments.

Derivation of the schemes. For $dS = a(S)dt + b(S)dW$, Euler-Maruyama is $S_{n+1} = S_n + a(S_n)h + b(S_n)\Delta W_n$. Scalar Milstein adds $\frac{1}{2} b(S_n)b'(S_n)[(\Delta W_n)^2 - h]$. Here $a(s)=\mu s$, $b(s)=\sigma s$, and $b'(s)=\sigma$:

$$S_{n+1}^{EM} = S_n^{EM}(1 + \mu h + \sigma \Delta W_n)$$

$$S_{n+1}^{Mil} = S_n^{Mil} \left[1 + \mu h + \sigma \Delta W_n + \frac{\sigma^2}{2} ((\Delta W_n)^2 - h) \right]$$

For reference, apply Itô's formula to $X_t = \log S_t$: $f(s)=1/s$ and $f''(s)=-1/s^2$. Thus $dX_t = (\mu - \sigma^2/2)dt + \sigma dW_t$. Integrating gives the exact path and exact step:

$$S_t = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right]$$

$$S_{t_{n+1}} = S_{t_n} \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) h + \sigma \Delta W_n \right]$$

For convergence tests, apply EM and Milstein to S itself. EM on $X = \log S$ reproduces the exact transition and has zero discretisation error.

1 · Strong and weak convergence

1.1 Strong error: compare two values from the same path

At final time, the exact solution on path m uses $W_t^{(m)} = \sum_n \Delta W_n^{(m)}$. For either method $A \in \{EM, Mil\}$, define its final error on the coupled path and then average its absolute value:

Repeated substitution of the update rules makes the terminal approximations explicit functions of the increments:

$$S_N^{EM} = S_0 \prod_{n=0}^{N-1} (1 + \mu h + \sigma \Delta W_n)$$

$$S_N^{Mil} = S_0 \prod_{n=0}^{N-1} \left[1 + \mu h + \sigma \Delta W_n + \frac{\sigma^2}{2} ((\Delta W_n)^2 - h) \right]$$

$$S_T^{(m)} = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) T + \sigma \sum_{n=0}^{N-1} \Delta W_n^{(m)} \right]$$

$$e_m^A = |S_N^{A, (m)} - S_T^{(m)}|, \quad e_{str}^A(h) = \mathbb{E}[e_m^A], \quad \hat{e}_{str}^A(h) = \frac{1}{M} \sum_{m=1}^M e_m^A$$

This is the exact definition of the strong error and its sample estimator; a closed elementary expression for $\mathbb{E}[S_N - S(T)]$ is not needed. Independently sampled exact paths would measure differences between Brownian paths as well as the method's time error.

The standard scalar-SDE orders here are $\mathbb{E}[S_N^{EM} - S(T)] = O(h^{1/2})$ and $\mathbb{E}[S_N^{Mil} - S(T)] = O(h)$. Thus $\log(\text{error})$ versus $\log(h)$ should have slopes near $1/2$ and 1 respectively. On the plot label the vertical axis estimated strong error.

The rate difference comes from the leading stochastic term omitted by EM, namely $\frac{1}{2}\sigma^2 S_n [(\Delta W_n)^2 - h]$. Milstein includes it. Under the standard regularity and finite-moment assumptions for GBM, the remaining errors accumulate at the stated global rates. These are asymptotic predictions to verify with several decreasing h values.

1.2 Weak error: take the expectation first

For a test function ϕ , the weak error is $|\mathbb{E}[\phi(S_N^A)] - \mathbb{E}[\phi(S(T))]|$. The pack requests the expectation of $S(T)$, so use $\phi(s)=s$. It follows that:

$$e_{wk}^A(h) = |\mathbb{E}[S_N^A] - \mathbb{E}[S_T]|, \quad \mathbb{E}[S_T] = S_0 e^{\mu T}$$

For EM, $\mathbb{E}[\Delta W_n] = 0$. For Milstein, $\mathbb{E}[(\Delta W_n)^2 - h] = 0$. Conditioning on the value at t_n therefore gives the same recurrence for both numerical means:

$$\mathbb{E}[S_{n+1}^A | S_n^A] = (1 + \mu h) S_n^A \implies \mathbb{E}[S_N^A] = S_0 (1 + \mu h)^N$$

$$\delta(h) = S_0 [(1 + \mu h)^{T/h} - e^{\mu T}], \quad e_{wk}(h) = |\delta(h)|$$

Provided $1 + \mu h > 0$ and $\mu \neq 0$, expand $\log(1 + \mu h) = \mu h - \mu^2 h^2 / 2 + O(h^3)$:

$$\delta(h) = -\frac{1}{2}S_0e^{\mu T}\mu^2Th + O(h^2) \implies e_{wk}(h) = O(h)$$

The two methods therefore have identical weak bias for $\phi(s)=s$, not merely the same expected slope. Their finite-sample estimates need not coincide. If $\mu = 0$, this particular weak bias is exactly zero at every h , so choose $\mu \neq 0$ or a different test function to observe a nonzero slope.

1.3 How to show Monte Carlo uncertainty

At each h , a direct estimate is $\delta = M^{-1}\sum S_n^{(m)} - S_0e^{\mu T}$. With sample standard deviation s of the M simulated terminal values, its standard error is $s/\sqrt{M}$; for sufficiently large M , a 95% interval for the signed bias is $\delta \pm 1.96s/\sqrt{M}$. The interval must be interpreted before taking the absolute value.

A sharper estimate exploits coupling. Compute $D_m = S_n^{(m)} - S_t^{(m)}$ using the same Brownian path; then $E[D_m] = \delta(h)$. Because the two values move together, the sample standard deviation s_D can be much smaller than s :

$$\hat{\delta}_{pair} = \bar{D}, \quad SE_{pair} = \frac{s_D}{\sqrt{M}}, \quad CI_{95\%}(\delta) = \bar{D} \pm 1.96 \frac{s_D}{\sqrt{M}}$$

With $s_D^2 = \sum (D_m - \bar{D})^2 / (M-1)$, increase M or restrict the fitted h range until the interval is narrow relative to $|\delta|$ and does not include zero. If an interval includes zero, plotting $|\delta|$ on a logarithmic axis can give a misleading slope. Check the fitted rate against the analytic $\delta(h)$ above; this GBM identity is a particularly strong validation of the Monte Carlo code.

Across several h values, summing adjacent fine-grid increments gives the coarse increment for the same path. Coupling different h values improves comparisons; it does not itself remove each estimator's sampling uncertainty.

2 · Derive and compare the terminal distribution

A standard Brownian motion has $W_t \sim N(0, T)$. The exact log-price is a constant plus a normal random variable:

$$\log S_T = \log S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T + \sigma W_T \sim N(m, v)$$

$$m = \log S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T, \quad v = \sigma^2 T$$

Exponentiating a normal variable gives a lognormal one. For $s > 0$ its exact terminal density and p -quantile are:

$$f_{S_T}(s) = \frac{1}{s\sqrt{2\pi v}} \exp\left[-\frac{(\log s - m)^2}{2v}\right], \quad s > 0$$

$$Q_{S_T}(p) = \exp\left[m + \sqrt{v} \Phi^{-1}(p)\right], \quad 0 < p < 1$$

For a chosen h and large M , plot a density-normalised histogram of terminal numerical values S_n^{EM} and/or S_n^{Mil} with $f_{S_t}(s)$ overlaid. Compare their empirical 5%, 25%, 50%, 75%, and 95% quantiles with $Q_{S_t}(p)$, in a table or quantile plot. The plotted sample should be the numerical terminal values: drawing only exact samples tests the random-number generator, not either numerical method.

On a coarse grid, numerical terminal values need not be exactly lognormal, even though the exact terminal variable is. EM or Milstein values can also be nonpositive; retain them when reporting positivity

and explain their treatment before drawing a density or quantile comparison.

3 • Exact mean and variance as sanity checks

For a normal $Y \sim N(0, T)$, its exponential moment is $E[e^{aY}] = e^{a^2 T/2}$. Apply it to W_t in the exact GBM solution:

$$\mathbb{E}[S_T] = S_0 e^{(\mu - \sigma^2/2)T} \mathbb{E}[e^{\sigma W_T}] = S_0 e^{\mu T}$$

Square S_t before taking expectations. The exponential-moment identity now uses $a = 2\sigma$:

$$\mathbb{E}[S_T^2] = S_0^2 e^{2(\mu - \sigma^2/2)T} e^{(2\sigma)^2 T/2} = S_0^2 e^{2\mu T + \sigma^2 T}$$

$$\text{Var}(S_T) = \mathbb{E}[S_T^2] - \mathbb{E}[S_T]^2 = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1)$$

For each scheme at a representative fine h , report its terminal sample mean and sample variance next to these exact values. State M and h . The exact-path simulation can be listed as an additional Monte Carlo check; finite samples are not expected to match the theoretical moments exactly. Mean accuracy alone does not establish distribution or variance accuracy.

4 • Optional: when positivity is lost

The exact solution is strictly positive when $S_0 > 0$. A numerical update multiplies its current value by a random factor F . Starting from a positive S_n , the next value is negative exactly when $F < 0$. Let $Z \sim N(0, 1)$ and $\sigma > 0$.

4.1 Euler-Maruyama

$$F_{EM}(Z) = 1 + \mu h + \sigma \sqrt{h} Z$$

$$F_{EM}(Z) < 0 \iff Z < -\frac{1 + \mu h}{\sigma \sqrt{h}}, \quad p_{EM}(h) = \Phi\left(-\frac{1 + \mu h}{\sigma \sqrt{h}}\right)$$

The normal distribution has an unbounded left tail, so $p_{EM}(h) > 0$ for every $h > 0$ when $\sigma > 0$. It is usually very small for fine steps if μ is fixed.

4.2 Milstein

$$F_{Mil}(Z) = \frac{\sigma^2 h}{2} Z^2 + \sigma \sqrt{h} Z + 1 + \left(\mu - \frac{\sigma^2}{2}\right)h$$

This is an upward-opening quadratic. Its vertex is $Z_* = -1/(\sigma\sqrt{h})$, with minimum $F_{\min} = [1 + (2\mu - \sigma^2)h]/2$. A negative factor has positive probability precisely when its minimum is strictly negative:

$$p_{Mil}(h) > 0 \iff (\sigma^2 - 2\mu)h > 1$$

When $(\sigma^2 - 2\mu)h > 1$, let $r = \sqrt{[(\sigma^2 - 2\mu)h - 1]}$. The factor is negative for Z strictly between z_- and z_+ :

$$z_{\pm} = \frac{-1 \pm r}{\sigma \sqrt{h}}, \quad p_{Mil}(h) = \Phi(z_+) - \Phi(z_-)$$

Otherwise $p_{Mil}(h) = 0$. At equality the factor reaches zero for one isolated value of Z , an event of probability zero. Milstein is therefore positive almost surely per step for the parameter range $(\sigma^2 - 2\mu)h \leq 1$, but it is not unconditionally positivity-preserving across all h and parameter choices.

4.3 What to measure

From $S_0 > 0$, record the fraction of M paths for which any step attains a negative value, at each h , separately for EM and Milstein. For independent update factors with per-step negative probability p , the chance of at least one negative factor is $1 - (1-p)^N$; the chance the terminal sign is negative is $[1 - (1-2p)^N]/2$. State which event the empirical fraction counts. A negative terminal value can distort a nonnegative terminal payoff calculated from that numerical value; no pricing analysis is required for the core GBM benchmark.

5 · Final experiment checklist and interpretation

Output	Recommended contents
Strong convergence figure	h on x-axis; $E S_n - S(T) $ on y-axis; both schemes, log-log axes, fitted slopes and path count.
Weak convergence figure	h versus $ E[S_n] - E[S(T)] $; both schemes; signed-bias estimates and 95% intervals; analytic bias as a check.
Distribution figure + quantiles	Numerical terminal histogram(s) with exact lognormal density; numerical versus theoretical quantiles.
Moments table	Sample mean and variance versus exact mean and variance, with h and M .
Positivity figure or table	Fraction with a negative intermediate value versus h ; compare observed fractions to update-factor formulas.

Limits of the GBM benchmark. Exact simulation exists here, so it is the reference for studying numerical methods, not a reason to prefer EM over the exact step for this model. In the project's next, non-affine model an exact solution is unavailable; the coupling, uncertainty, and validation habits established here become especially valuable. If $\sigma = 0$, the problem is deterministic and the stochastic orders and positivity probabilities above need a separate interpretation.

Source. Project materials supplied for MTH321: Problem Pack: Five Suggested Directions for the ODE IVP Project, Direction ④, pp. 6–7; and ODE IVP Project Brief, topic ④. The derivations and experiment design in this guide expand those requirements.